Denis Tolkovets

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EDUCATION

BSc Mechanical Engineering – Product Engineering and Context (MPEC)
10/2022 – 08/2027 (Expected) | TH Köln – University of Applied Sciences, Germany

- BSc thesis 2027 (Furhat Robotics): asynchronous dual-LLM architecture for conversational HRI.
- Problem-Based Learning (PBL) with smart mechatronic systems and product development.
- Focus on research methodology: projects concluded with empirical evaluations and scientific papers.
- Elective specialisation: Social Robotics (Human-Machine-Interaction, Service and Industry Robotics).
- Current GPA: 1.8 (German scale: 1.0 best, 4.0 pass).
- Strong results in research courses: Product Engineering I & II (1.0) and Social Robotics (1.7).

RESEARCH PROJECTS

Open-Source Smart Glasses Kit
10/2024 – 05/2025 | Course: Product Engineering II

- Engineered a low-cost, MLLM-based smart glasses kit for easy replication by non-experts.
- Designed hardware and software to enable assembly without special tools or programming.
- Conducted a usability study where 5 out of 6 non-experts assembled and operated the kit unaided.

Voice-Controlled Cobot System
10/2024 – 02/2025 | Course: Service and Industry Robotics

- Developed a ROS 2 service-based architecture with a central manager orchestrating dialogue, object detection, and robotic arm control for collaborative tool fetching and stowing.
- Linked STT, LLM intent recognition, and TTS for hands-free interaction in a shared workspace.
- Derived closed-form IK for a 5-DOF arm, bypassing 6D solver limits caused by coupled orientation.
- Iteratively trained a YOLO11-OBB model for oriented tool detection, raising mAP50-95 to ~81%.

Autonomous Harvesting Robot
10/2023 – 10/2024 | Course: Engineering Office III & Product Engineering I

- Developed an autonomous harvesting system, combining a rover, a robotic arm, and a detector.
- Modelled kinematics and implemented P/PI control for approaching and picking up apples.
- Built a ToF-based 2D scanner, trained a YOLOv8 model, and compared detectors quantitatively.
- Demonstrated the YOLO-based system was ~18× faster and reduced task error rate from 40% to 0%.

Retrieval-Augmented Generation (RAG) Study Chatbot
02/2024 – 10/2024 | Course: Human-Machine-Interaction

- Built a RAG chatbot for exam preparation with an interface that displays information sources.
- Quantitatively compared retrieval-augmented GPT-4o with base GPT-4o on hallucination rates.
- Conducted a usability study to assess the helpfulness of the displayed information sources.

WORK EXPERIENCE

Prototyping & Concepting Intern – Furhat Robotics AB
07/2025 – 04/2026 | Stockholm, Sweden (full-time)

- Developed a local real-time object detection module, including a custom interface for dynamic parameter tuning (confidence threshold, processing resolution, class filtering).
- Built full-stack prototypes using WebSockets and n8n to enable automated conversation summarisation and long-term memory for social interaction.
- Conducted electromechanical troubleshooting, including PCB flashing and soldering, to resolve display and servo motor faults, and documented root causes for the engineering team.
- Prototyped a pipeline using generative models with multimodal inputs (text and image) to produce face textures compliant with the robot's projection geometry.

Student Teaching Assistant – Institute of Automotive Engineering (TH Köln)
10/2024 – 06/2026 | Cologne, Germany (Remote since 07/2025, part-time)

- Tutored students in embedded C++ for mobile robots, assisting with PID control, sensor fusion (IMU/ToF/Ultrasonic), and hardware-software integration.
- Guided students in designing algorithms for autonomous tasks like maze navigation, parking, and collision avoidance, providing feedback on logic and code structure.
- Developed study materials for sensor data visualisation and Git workflows.
- Graded scientific reports, controller documentation, and UML system models.

Operations & Production – Serverhero GmbH
01/2018 – 09/2024 | Cologne, Germany (full-time & part-time)

- Supported server production through hardware assembly, testing, and quality assurance.
- Developed Excel tools (VBA, pivot tables) to automate accounting processes.

SKILLS

- **Programming & Tools:** Python, C++, MATLAB/Simulink, Linux, WSL, Docker, Git, CMake, Conda
- **Robotics:** ROS 2 (Galactic), MoveIt 2, Ignition Gazebo, RViz2, TF2
- **AI:** Computer Vision (YOLOv8/11, Ultralytics/PyTorch), RAG, Multimodal LLMs, Speech (STT/TTS)
- **Electronics:** ESP32, Arduino, Raspberry Pi, Sensor Integration, Circuit Prototyping, Soldering
- **Mechanical:** Autodesk Fusion 360 (CAD), Finite Element Analysis, FDM and Resin 3D Printing
- **Evaluation:** Usability Studies, Statistical Analysis (*t*-test, *z*-test, *F*-tests, Fisher's exact test)
- **Collaboration:** Agile Project Management, Structured Team Reflection
- **Languages:** German (Native), English (C1, IELTS 7.5)

VOLUNTEER

Erasmus Ambassador
04/2026 – Present | Remote

- Guidance and support for subsequent exchange students.

Finance Administrator – Bahn-Landwirtschaft e.V.
08/2022 – Present | Cologne, Germany

- Manage allotment garden finances, maintain financial spreadsheets, and issue member invoices.